# Towards an fNIRS foundation model using self-supervision to improve machine learning classification

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**Abstract:** We propose a novel approach based on self-supervised learning taking advantage of unlabelled segments of functional near-infrared spectroscopy (fNIRS) recordings to pretrain a neural network for classifying *n*-back levels from brain data. A self-supervised convolutional encoder-decoder was trained on a pretext task consisting in predicting deoxyhaemoglobin (HbR) data from oxyhaemoglobin (HbO), and vice versa. The trained encoder can then be used as a basis for training a subsequent supervised classifier. Ongoing work is investigating this self-supervised approach with more unlabelled data from other datasets to build a robust fNIRS foundation model.

**Introduction:** While machine learning models have been widely used to classify functional near-infrared spectroscopy (fNIRS) data, they often have limited generalisation capabilities[1]. An explanation may lie in the limited size of available datasets combined with the high dimensionality of fNIRS recordings (multi-channel time series data). Data augmentation can be used to mitigate this issue, however, since it is common to have in fNIRS recordings unlabelled intervals separating the task trials or other tasks unrelated to the classification goal, we investigated an approach to make the most out of the real data at disposition instead of creating artificial data. It consisted in using unlabelled fNIRS recording segments for a self-supervised representation learning task to pretrain a machine learning classifier.

**Methods:** We focused in this work on a dataset of *n*-back tasks[2], a common paradigm for studying mental workload, with the goal of performing classification of the different conditions (0-back vs. 2-back vs. 3-back) in a subject-independent fashion (evaluation with leave-subjects-out). For this purpose, we first used a pretext task consisting in reconstructing deoxyhaemoglobin (HbR) signal from oxyhaemoglobin (HbO) and HbO signal from HbR, performed with a convolutional encoder-decoder neural network model (*Fig. 1*). The model therefore encoded each channel type into features based on the relationship with the other channel type. This pretext task was performed using all the labelled *n*-back data as well as the remaining data of the recordings (intervals between tasks) in a self-supervised manner. After training the model on this pretext task, this learnt knowledge is transferred to the downstream classification task, by connecting the encoder's pretrained layers to fully connected classification layers that were subsequently trained on the labelled *n*-back data only, in a supervised manner. The models were trained, optimised and evaluated with *BenchNIRS*[1] and come to extend this open-source Python benchmarking framework, adding support for self-supervised learning with fNIRS.

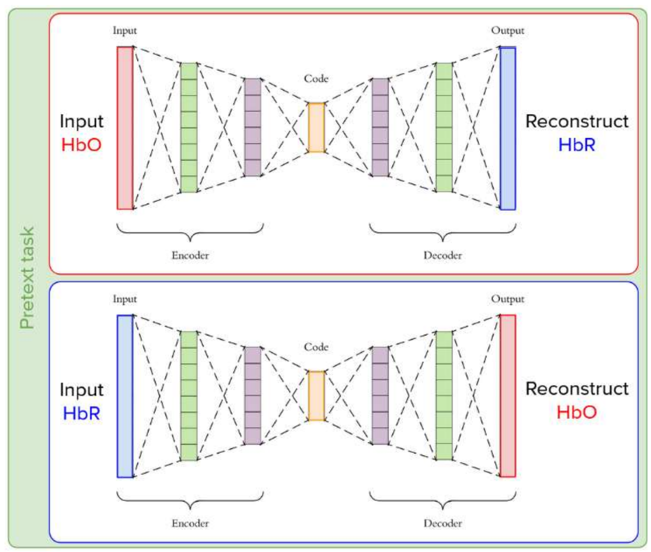


*Fig. 1: Self-supervised haemoglobin signal reconstruction*

**Results:** This approach enabled us to train the model on 104 extra unlabelled trials in addition to the 702 labelled trials. Initial results when pretraining our model on the reconstruction task enabled to reach a classification accuracy of 34.8% as evaluated on data from unseen subjects with 5-fold cross validation, which is low but on par with subject-independent fully supervised classification baseline benchmarks on this *n*-back task which remains challenging[1].

**Conclusion:** Our explorations are opening new avenues for future research in machine learning with fNIRS, since using pretext tasks enables to introduce expert knowledge into models to go towards more relevant and explainable inferences. Ongoing work is focusing on improving those initial results by performing the self-supervised learning task on much more unlabelled data from other datasets with the goal of building a robust foundation model for fNIRS.

*1. J Benerradi et al., 2023 - 2. J Shin et al., 2018*